

# Can Artificial Intelligence Be Added to an Inspection System, or Does It Have to Be Built In?



*It is the question the electronics assembly industry has largely avoided asking out loud. Delvitech, the Swiss company that built its AOI and SPI platforms AI-native from day one, argues that the answer determines everything that follows: false-call rates, coverage of atypical components, and the true cost of owning inspection equipment.*

## A retrofit is not an architecture

Almost every supplier of automated optical inspection now describes its products as AI-powered, and in most cases the description is accurate. A neural network is present. It does useful work. What the phrase rarely reveals is where that network sits.

In the majority of systems on the market, the network was added to a decision architecture designed years earlier around rule-based thresholds. Deterministic algorithms measure a solder joint against tolerance windows and raise a flag; the network is then asked whether the flag deserves to be trusted. This is a genuine improvement, and it produces a measurable drop in false calls. But the primary judgement is still made by rules, and the network never sees the evidence that produced them. It is a second opinion on a verdict already reached.

Delvitech began from the opposite end. Founded in Rancate, Switzerland, in 2018, the company specified its inspection platforms around proprietary neural networks from the first drawing rather than adding them to an existing product line. Horus, its all-in-one platform, performs AI 3D solder paste inspection together with pre-reflow and post-reflow AOI in true 3D. Aton addresses demanding post-reflow work, with multilevel inspection up to 140 mm. In neither case does the network act as a filter above a rule engine. It is the decision layer.

## Where the intelligence sits determines what is permitted to see

Delvitech specified the sensor to feed the network. Its patented optical head uses phase shift profilometry, projecting sequential structured-light patterns through four synchronised projectors at 90-degree phase offsets. A single projector set measures features from a few micrometres up to 30 mm in height, which allows solder paste deposits and tall connectors to be measured on the same hardware. A working distance of roughly 300 mm keeps low-profile features adjacent to tall components measurable rather than shadowed. High-resolution 2D imagery and true 3D measurement are fused rather than sequenced, with the platform processing over 40 Gbit of inspection data per second, and a 50 by 50 mm field of view inspected in 0.6 seconds at 2 micrometer height resolution.

## The last two per cent is where the cost concentrates

Contemporary inspection technology handles most standard production scenarios automatically. This is where retrofitted intelligence struggles most.

**Delvitech's answer is the Training Manager**, which relocates the training lifecycle to the people who understand the process. Some training modes complete in under a minute. A single accurately labelled image can measurably improve model performance, and the tool is integrated into offline programming stations so that models are developed in parallel with live production and deployed in-line once validated, without pausing the line. Operators can annotate and train it on-site to recognize and assess non-standard elements it initially cannot detect.

## The problem is that the industry does not advertise

Any network that learns continuously faces a failure mode documented in the machine learning literature since 2013: catastrophic forgetting, in which a model trained on new data loses information it had already acquired.

**Delvitech's Knowledge Retention Technology is a continual learning method** built to mitigate it. New image datasets can be absorbed without compromising previously acquired knowledge, allowing multiple models to evolve progressively rather than being rebuilt.

That knowledge is then portable. Validated models are distributed through a central library across every platform on a line, throughout a facility, and where required across geographically distributed plants, standardising inspection criteria and removing duplicated training effort between sites.

## What the architecture pays back

The industry evidence for AI-enabled inspection is already substantial.

Delvitech's claim is a behaviour, not a benchmark: because the neural networks interpret components and process variation rather than comparing them against static thresholds, false calls fall by a factor of ten. And keep falling, because every operator validation feeds the model. How far they fall on a given line is something Delvitech prefers

to demonstrate on the customer's own boards rather than assert in print.

The consolidation argument matters to finance: one platform for SPI and AOI means fewer spares, simpler training, faster setup and lower total cost of ownership. Fewer false calls also reduce rework, scrap and waste.

Future proofing comes from architecture, not promises. Self-programming and predictive functions can be added to an AI-native platform; retrofitting them onto threshold-based systems requires a redesign.

## Why the question matters here, and now

India's rapidly scaling, high-mix electronics industry needs adaptable inspection systems that evolve with products and production demands.

## Made in Bangalore, to Swiss standards

Delvitech, recognized with the Swiss Economic Forum's SEF.Growth High-Potential Label in 2025, is opening a manufacturing facility in Bangalore, alongside its Swiss headquarters and its US operation. At its core is a machine shop equipped with latest-generation high-precision CNC machining and inspection, producing critical components to the same standards as in Switzerland. Machines will be manufactured and assembled locally, with end-to-end testing, shortening lead times and delivery costs for Indian customers.

The facility will support customer trials, evaluations, training and application support using customers' own boards on a Delvitech platform and watch. It is a long-term commitment to India, in the spirit of "Make in India".

The decision facing a manufacturer investing in inspection today is not simply which system detects most accurately on the boards currently in production. It is which one will still be learning in five years, and who will own the knowledge it has acquired by then.

**Delvitech will be at productronica India in Bangalore, 16–18 September, Hall 4 - Booth B21. The simplest way to settle the question this article opened with is to bring your most difficult board.**

## SOURCES:

- Inspection-time and product-return figures: Intelgic, Quality Inspection Trends in Manufacturing Using AI (2025); Data Bridge Market Research, Global Automated Optical Inspection Market Report.
- Catastrophic forgetting: Goodfellow, I. et al., An Empirical Investigation of Catastrophic Forgetting in Gradient-Based Neural Networks (2013), arXiv:1312.6211.
- Platform specifications, Training Manager functionality and application cases: Delvitech technical documentation and the Delvitech Training Manager white paper.
- Specifications may vary depending on system configuration. Self-programming and predictive functions are scheduled from 2026/early 2027.